

Vaaheesan S

Software Engineer

C++ • Python • Software Engineering • Backend Systems

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Portfolio GitHub LinkedIn LeetCode HackerRank SkillRack

Professional Experience

Backend Lead Intern – Hexaware Technologies *Feb 2026 – Present*

- Led a 3-member backend sub-team within a 12-member engineering team, coordinating feature implementation, backend integration, and onboarding new developers.
- Designed and implemented RESTful APIs, schema-driven backend services, and database workflows using Python and FastAPI, supporting applications serving supporting applications serving 3,000 concurrent users.
- Developed asynchronous AI evaluation pipelines using Redis queues and `asyncio`, integrated Alembic for version-controlled database migrations, and maintained code quality using Ruff.
- Contributed to deployment validation, CI/CD workflows, production monitoring, and backend optimization, reducing application performance overhead by approximately 3%.

Web Developer Intern – Infotak Solutions *Aug 2025 – Jan 2026*

- Developed and deployed 8+ full-stack applications, including an e-commerce platform and an IoT-integrated management system, working across frontend, backend, and database components.
- Identified database performance bottlenecks during testing of the e-commerce platform, analyzed SQL query execution, and improved application response time through indexing and query restructuring.
- Improved frontend performance and accessibility by implementing lazy loading, responsive UI design, and WCAG-compliant interfaces, reducing load time across multiple projects.

Software Development Intern – J7 Technology Solutions Pvt. Ltd. *Aug 2025 – Sept 2025*

- Developed student, teacher, and parent-facing modules for a cloud-based School CRM platform using Next.js, Node.js, Express.js, MongoDB, and AWS.
- Collaborated within a 5-member engineering team to build scalable backend APIs, integrate role-based functionality, and deliver production-ready modules across multiple user roles.

Projects

School CRM Platform – Full-Scale Digital School Management System *(Private Codebase)* *(Next.js, Node.js, Express.js, MongoDB, AWS)*

- Contributed to the development of a full-scale school management platform supporting seven user roles including students, teachers, parents, HR, principal, support staff, and finance.
- Designed and implemented student, parent, and teacher modules featuring attendance management, academic records, communication workflows, and administrative operations.
- Built scalable backend APIs with Node.js and Express.js, integrated MongoDB for data management, and deployed the application on AWS.

Spendly – Personal Expense Tracker *(React 18, Node.js, Express.js, MongoDB)*

- Built a full-stack expense management platform supporting expense tracking, category-based budgeting, analytics dashboards, and spending insights.
- Implemented JWT authentication, Google OAuth integration, secure password reset using Nodemailer, and role-based authentication workflows.
- Designed secure backend APIs with validation, Helmet security middleware, rate limiting, pagination, filtering, sorting, and search capabilities.
- Built a responsive React frontend using Context API for centralized state management, enabling seamless CRUD operations and interactive dashboards.

Elevatr – Job and Internship Portal *(Next.js, TypeScript, FastAPI, PostgreSQL)*

- Built a two-sided recruitment platform enabling students to discover opportunities, bookmark jobs, apply online, and monitor application status while recruiters manage hiring pipelines.
- Implemented recruiter dashboards with real-time statistics, application tracking, optimistic UI updates, and role-based authorization.

- Designed and developed a FastAPI backend with PostgreSQL, implementing secure PBKDF2-SHA256 password hashing, JWT authentication, and role-based access control.
- Developed the frontend using Next.js and TypeScript, resolved API routing inconsistencies, and Maintained a modular and production-ready codebase following ESLint best practices.

MindSync+ – AI-Assisted Mental Wellness Platform (MSME 5.0 Funded) *(Next.js, FastAPI, PostgreSQL, LLMs)*

- Conceptualizing an AI-assisted mental wellness platform that aims to connect users with peer supporters while providing an intelligent conversational fallback when volunteers are unavailable.
- Designing the backend architecture using FastAPI and PostgreSQL to support secure authentication, conversation workflows, and scalable request handling.
- Exploring LLM-assisted conversational support with moderation and escalation mechanisms to deliver context-aware and empathetic interactions.
- Selected under the MSME 5.0 Government Funding Scheme for its innovative approach to AI-assisted mental health support.

Technical Expertise

Programming Languages	C++ Python Java SQL JavaScript TypeScript
Backend & APIs	FastAPI Flask Express.js Node.js REST APIs
Frontend	React.js Next.js HTML5 CSS3 Tailwind CSS Vite
Databases	PostgreSQL MySQL MongoDB SQLite
AI/GenAI	LLM LangChain Azure OpenAI LangSmith Prompt Engineering
Cloud/DevOps	Docker AWS Git CI/CD
Tools	Postman VS Code GitHub Redis Alembic Ruff

Competitive Programming

- **LeetCode** – Solved 131+ coding problems with a Contest Rating of **1655** (Top **17.5%**).
- **SkillRack** – Solved 308+ programming problems, Ranked #1 in College Programming Test Leaderboard, earned certifications in Data Structures (C) and SQL Basics.
- Regularly practice algorithmic problem solving through LeetCode, HackerRank, and SkillRack.

Achievements

- Selected for the MSME 5.0 Government Funding Scheme for the project **MindSync+** (2026)
- 1st Place – QR Debugging Event (Namma VEC 2026)
- 2nd Place – App Development (Namma VEC 2025)
- Participant – Cognizant Technoverse Hackathon 2026
- Participant – VISA 24 Hours AI Hackathon (IITM, 2025)

Certifications

- NPTEL – Programming in Modern C++
- NPTEL – Programming in Java
- NPTEL – Deep Learning
- NPTEL – Natural Language Processing
- NPTEL – Introduction to Internet of Things
- Udemy – Complete Python Course

Education

B.E. Computer Science and Engineering *2023 – 2027*
Velammal Engineering College
CGPA: **8.33**

AISSCE – Class XII *2022 – 2023*
Velammal Vidhyashram
68.6%

AISSE – Class X *2020 – 2021*
Velammal Vidhyashram
81%

Interests

Traveling • Badminton • Technology Exploration